

EXCEL TRAINING SESSIONS

Ship Mode

Delivery Truck

Express Air

Regular Air

Sum of Order Amount

Column Labels

Row Labels

Express Air

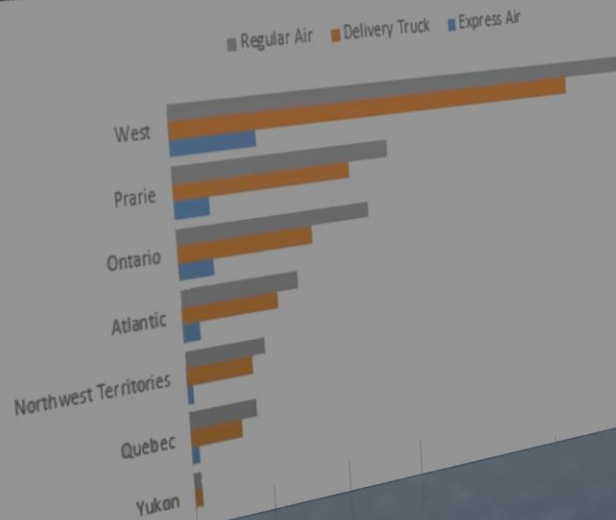
Delivery Truck

Regular Air

Grand Total

Yukon	3,496	51,467	51,842	106,805
Quebec	49,165	326,427	422,010	797,602
Northwest Territories	39,225	412,971	498,952	951,147
Atlantic	108,753	602,160	737,511	1,448,424
Ontario	219,227	853,402	1,232,791	2,305,420
Prarie	222,857	1,124,684	1,391,724	2,739,265
West	534,139	2,713,097	3,144,607	6,391,843
Grand Total	1,176,861	6,084,208	7,479,437	14,740,506

Regular Air Delivery Truck Express Air



• PIVOT TABLE EXERCISE

Pivot table from a single table

Open the PP_SampleData_START.xlsx file

Go to the Invoices sheet and click inside the table.

In the Table Design Menu, choose Summarize with Pivot table then choose 'New worksheet'.

The screenshot shows the Excel interface with the 'Table Design' menu open. The 'Summarize with PivotTable' button is highlighted with a green callout arrow labeled '1'. Below the menu, a table of invoice data is visible. To the right, the 'PivotTable from table or range' dialog box is open, showing the 'Table/Range' as 'tbl_commandes' and the 'New Worksheet' option selected. A green callout arrow labeled '2' points to the 'New Worksheet' radio button.

Table Name:

Table/Range:

Choose where you want the PivotTable to be placed

☒ New Worksheet

☐ Existing Worksheet

Location:

Choose whether you want to analyze multiple tables

☐ Add this data to the Data Model

OK Cancel

lineNB	invoiceID	clientID	serviceID	NBHoursSol	date
1	20-0001	76	32	117	02.01.2020
2	20-0002	141	36	24	03.01.2020
3	20-0003	238	28	2	04.01.2020
4	20-0004	183	22	116	05.01.2020
5	20-0005	246	33	120	06.01.2020

Pivot table from a single table

In the empty pivot table worksheet, use the right pane to set up the pivot table by dragging the NBHoursSold field to the values, Date to columns and serviceID to rows.

Sum of Column				
	2020	2021	2022	Grand Total
Row				
1		428	642	1070
2		628	638	1266
3		970	1232	2202
4		1050	340	1390
5		1090	1350	2440
6		1388	1554	2942
7		710	196	906
8		418	368	786
9		486	586	1072
10	612	556	440	1608
11	314	496	602	1412
12	476	500	272	1248
13	670	952	836	2458
14	1050	370	898	2318
15	698	1268	1478	3444
16	298	112	348	758
17	614	1150	1026	2790
18	994	758	928	2680

PivotTable Fields

Choose fields to add to report:

Search

☐ lineNB

☐ invoiceID

☐ clientID

☒ serviceID

☒ NBHoursSold

☒ date

☒ Months (date)

☒ Quarters (date)

☒ Years (date)

More Tables...

Drag fields between areas below:

Filters

Columns

Rows

Values

Years (date)

Quarters (date)

Months (date)

date

serviceID

Sum of NBHoursSold

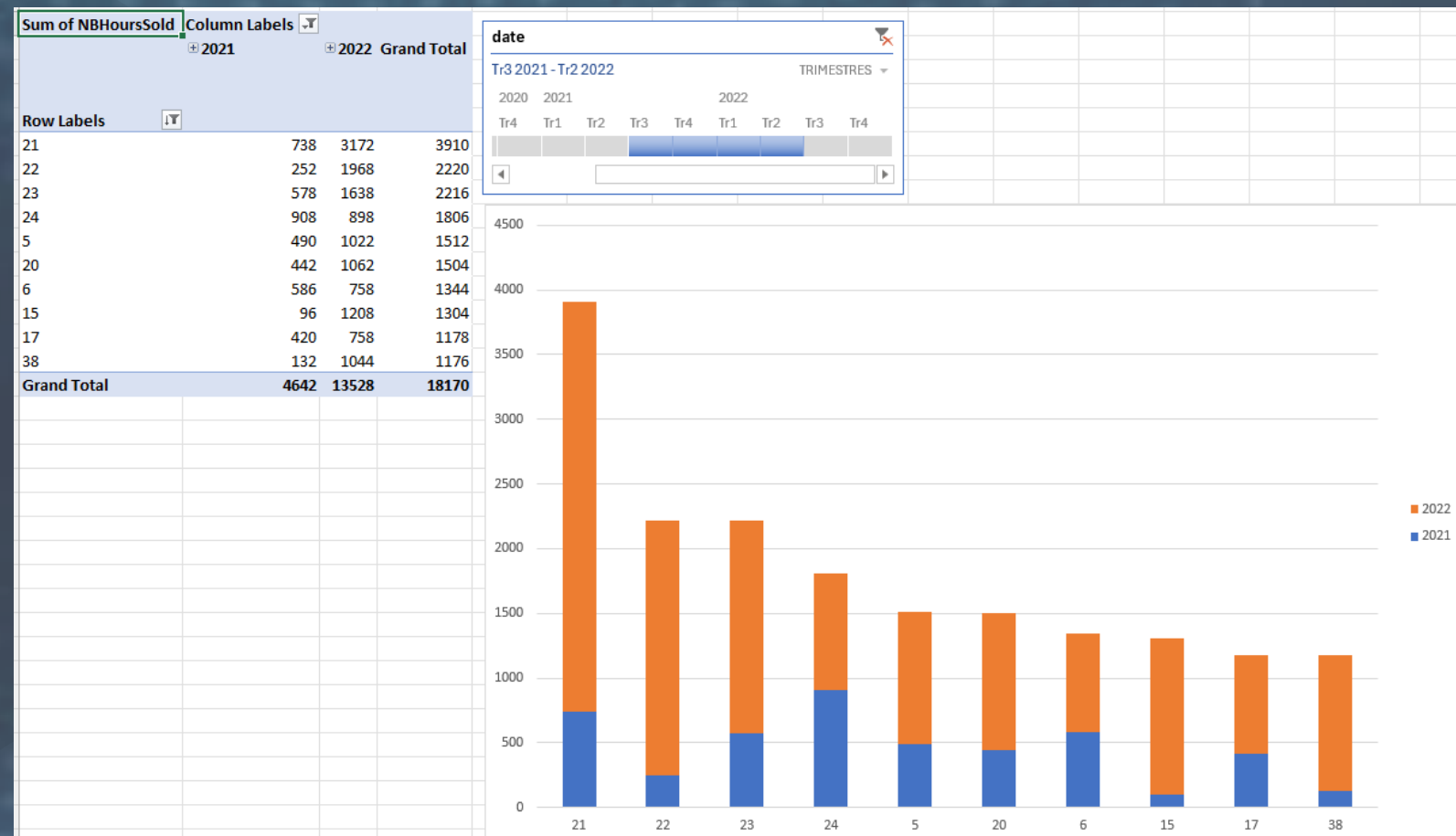
Insert a Pivot chart and a slicer

Insert a pivot chart from the pivot table (stacked columns).

Insert a timeline slicer with the date.

Sort the pivot table by the NBHoursSold measure in descending order.

Filter the top 10 values in the sorted table.



Get & prepare the Data from multiple sheets

Open the PP_SampleData_START.xlsx file

From the Services sheet, click inside the table and go to Data > From Table/Range.

The table is loaded into the Power Query editor.

Repeat the process for the Cat and Subcat tables.

File

Home

Insert

Page Layout

Formulas

Data

Review

View

Developer

Power

Get Data

From Text/CSV

From Web

From Table/Range

From Picture

Recent Sources

Existing Connections

Get & Transform Data

Refresh All

Queries & Connections

Properties

Workbook Links

Queries & Connections

C5

✕

✓

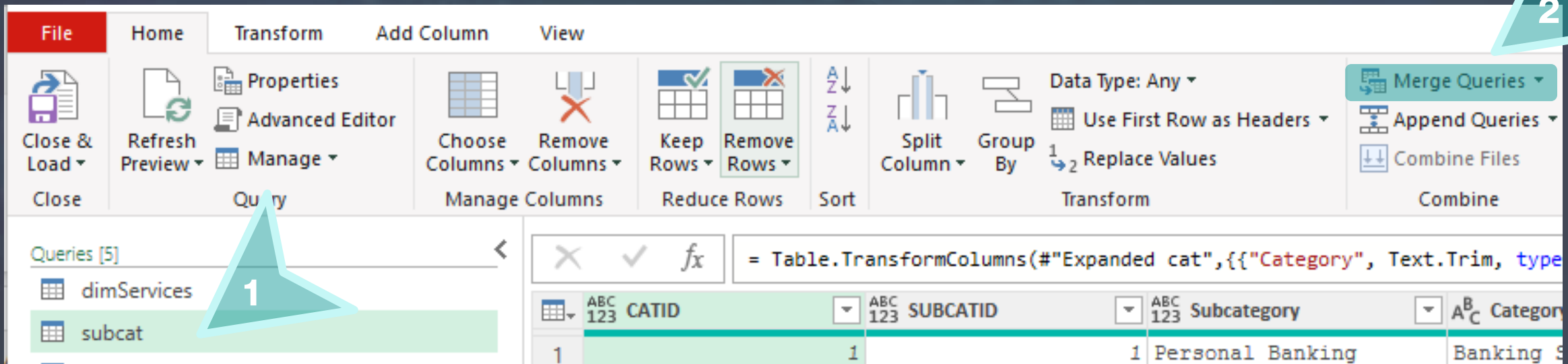
fx

BKS-4

	A	B	C	D
1	lineNE	Subcategory	Reference	Description
2	1	1	BKS-1	Savings Accounts
3	2	1	BKS-2	Checking Accounts
4	3	2	BKS-3	Business Loans
5	4	2	BKS-4	Merchant Services
6	5	3	IS-1	Portfolio Diversification
7	6	3	IS-2	Asset Allocation
8	7	4	IS-3	Equity Trading
9	8	4	IS-4	Options Trading
10	9	5	RP-1	Retirement plans
11	10	5	RP-2	Retirement plans (Roth IRA)

Use Power Query to merge tables

In Power Query editor, select the Subcat query and Click on Home > Merge Queries.



The screenshot shows the Power Query editor interface. The 'Home' ribbon is active, displaying various transformation options. The 'Queries [5]' list on the left contains 'dimServices' and 'subcat', with 'subcat' selected. The main area shows the formula bar with the M code: `= Table.TransformColumns(#"Expanded cat",{{"Category", Text.Trim, type`. Below the formula bar, a preview of the data is shown, including columns for CATID, SUBCATID, Subcategory, and Category.

ABC 123	CATID	ABC 123	SUBCATID	ABC 123	Subcategory	ABC 123	Category
1		1		1	Personal Banking		Banking S

Use Power Query to merge tables

Select the Cat table as the second table in the popup window then select the common column in both tables.

Verify that Matches are correctly made.

Click OK.

The screenshot shows the 'Merge' dialog box in Power Query. It contains two tables: 'subcat' and 'cat'. A dropdown menu shows 'cat' selected. Below the tables, the 'Join Kind' is set to 'Left Outer (all from first, matching from second)'. There is a checkbox for 'Use fuzzy matching to perform the merge' which is unchecked. A section for 'Fuzzy matching options' is collapsed. At the bottom, a status bar indicates 'The selection matches 20 of 20 rows from the first table.' and an 'OK' button is present.

2

Select a table and matching columns to create a merged table.

subcat

CATID	SUBCATID	Subcategory	Category
1	1	Personal Banking	Banking Services
1	2	Business Banking	Banking Services
2	3	Investment Management	Investment Services
2	4	Stock Trading	Investment Services
3	5	Retirement Accounts	Retirement Planning

cat

catid	Category
1	Banking Services
10	Financial Technology
2	Investment Services
3	Retirement Planning
4	Tax Services

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

> Fuzzy matching options

✓ The selection matches 20 of 20 rows from the first table.

OK

1

2

3

4

Expand merged table in query Subcat

ABC 123	CATID	ABC 123	SUBCATID	ABC 123	Subcategory	cat
1		1		1	Personal Banking	Table
2		1		2	Business Banking	Table
3		2		3	Investment Management	Table
4		2		4	Stock Trading	Table
5		3		5	Retirement Accounts	Table
6		3		6	Annuities	Table
7		4		7	Individual Tax	Table
8		4		8	Business Tax	Table
9		5		9	Life Insurance	Table
10		5		10	Property & Casualty	Table
11		6		11	Wills & Trusts	Table
12		6		12	Inheritance Planning	Table
13		7		13	Financial Consulting	Table
14		7		14	Investment Advisory	Table
15		8		15	Home Loans	Table
16		8		16	Commercial Mortgages	Table
17		9		17	Credit Counseling	Table
18		9		18	Debt Management	Table
19		10		19	Online Banking	Table
20		10		20	Blockchain & Cryptocu...	Table

ABC 123 Subcategory cat

Search Columns to Expand

☒ Expand ☐ Aggregate

☐ (Select All Columns)

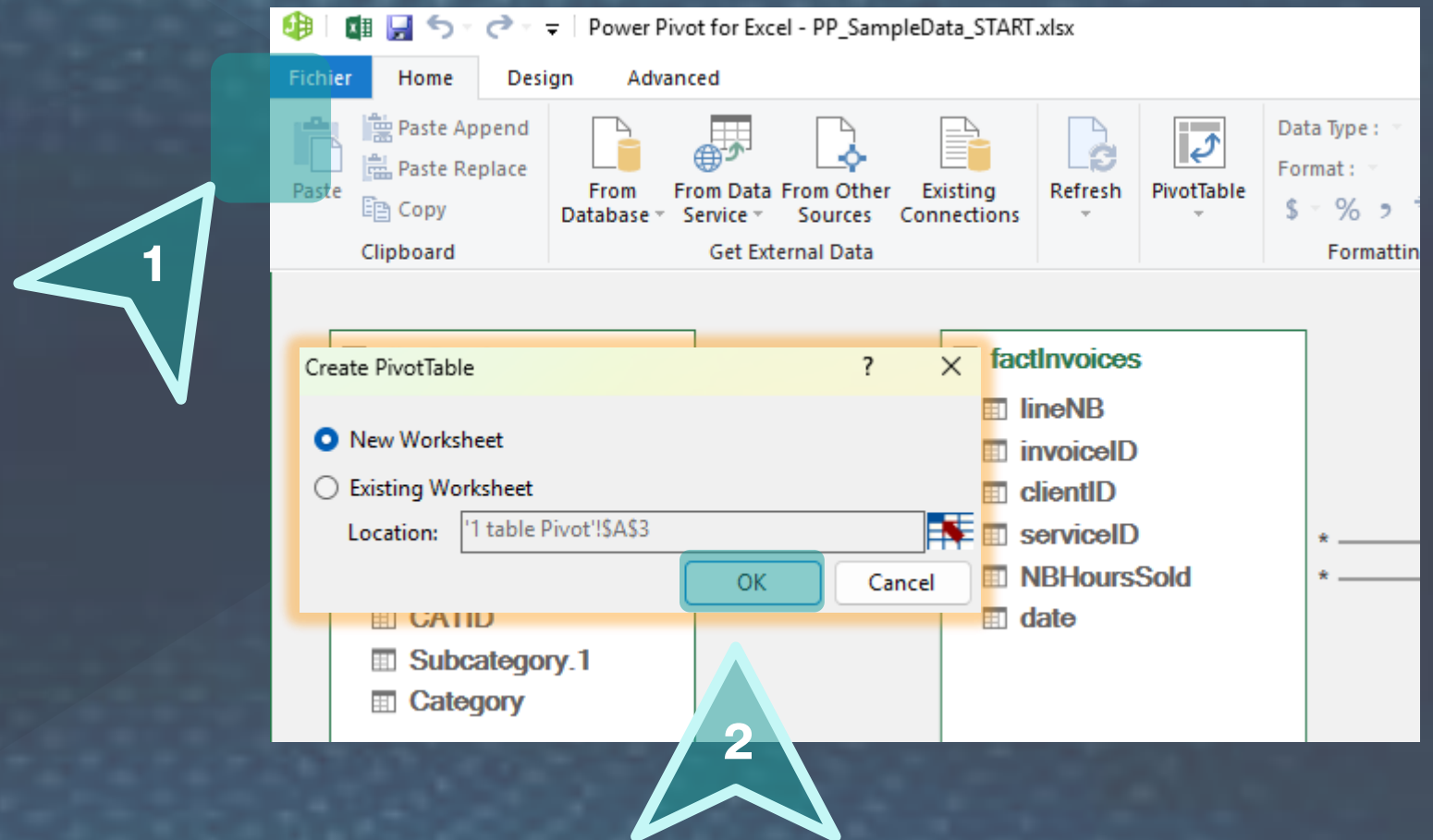
☐ catid

☒ Category

☐ Use original column name as prefix

OK Cancel

Import data from the CLIENTS table



Create flat file

Duplicate the cat query and change the name of the table in the source step to the name of the table in the CLIENTS sheet. Rename to dimClients.

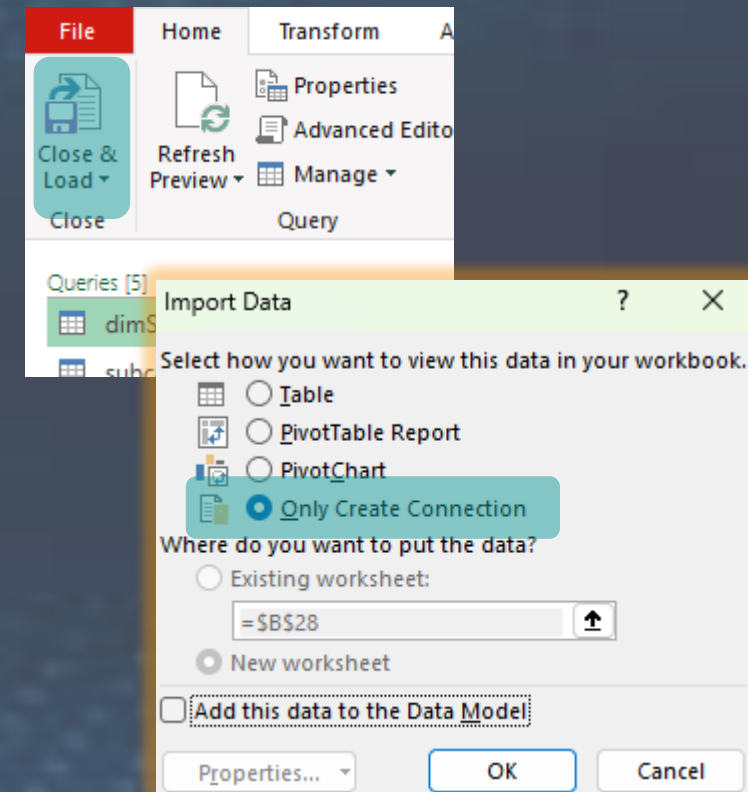
Repeat the same procedure for the INVOICES table and rename the Invoices query to factInvoices.

Merge the factInvoices and dimClients queries and add the client name, type, location and canton columns to the fact table.

Merge factInvoices and dimServices queries and add the Services Description, rate per hour, category and subcategory columns to the fact table. Rename the columns for the names as the cat and subcat.

factInvoices is now ready to be summarized into a pivot table.

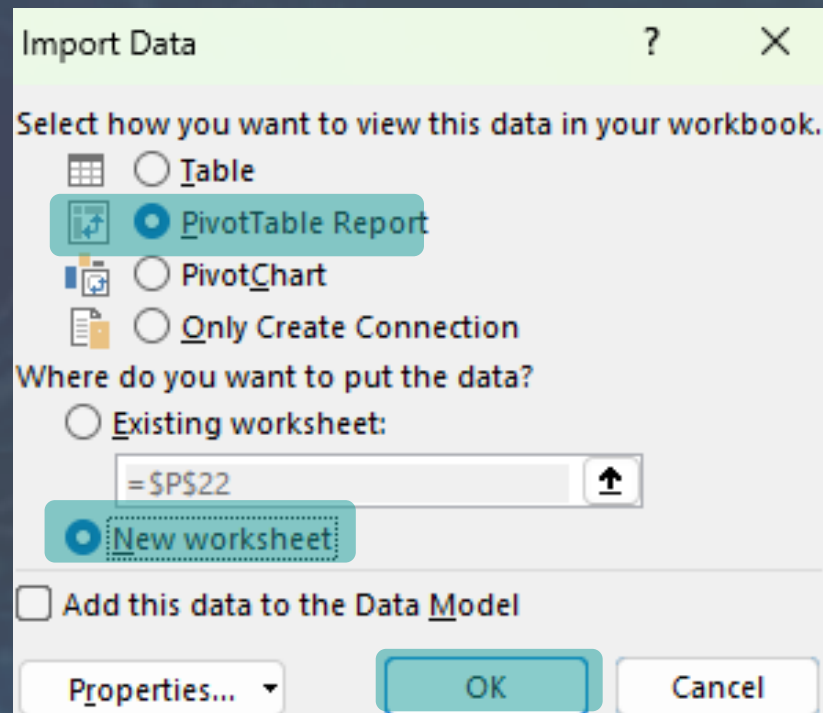
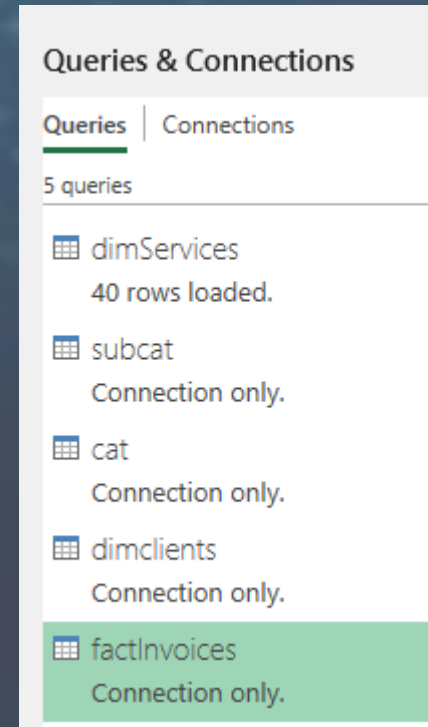
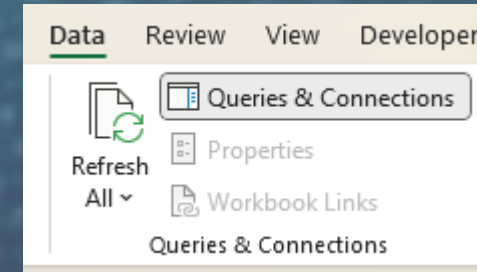
Click on Home > Close & Load > Only create connection



Create Pivot Table from the factInvoices query

In excel, Go to Data > Queries & Connections.

In the right pane that appears, right click on the factInvoices Query and select 'Load to' and choose the following options:



A new Pivot table is created

All the fields from the different merges are now available. Drag the NBHoursSold in the values window and the canton in the rows. You can rename the column in the pivot table if desired.

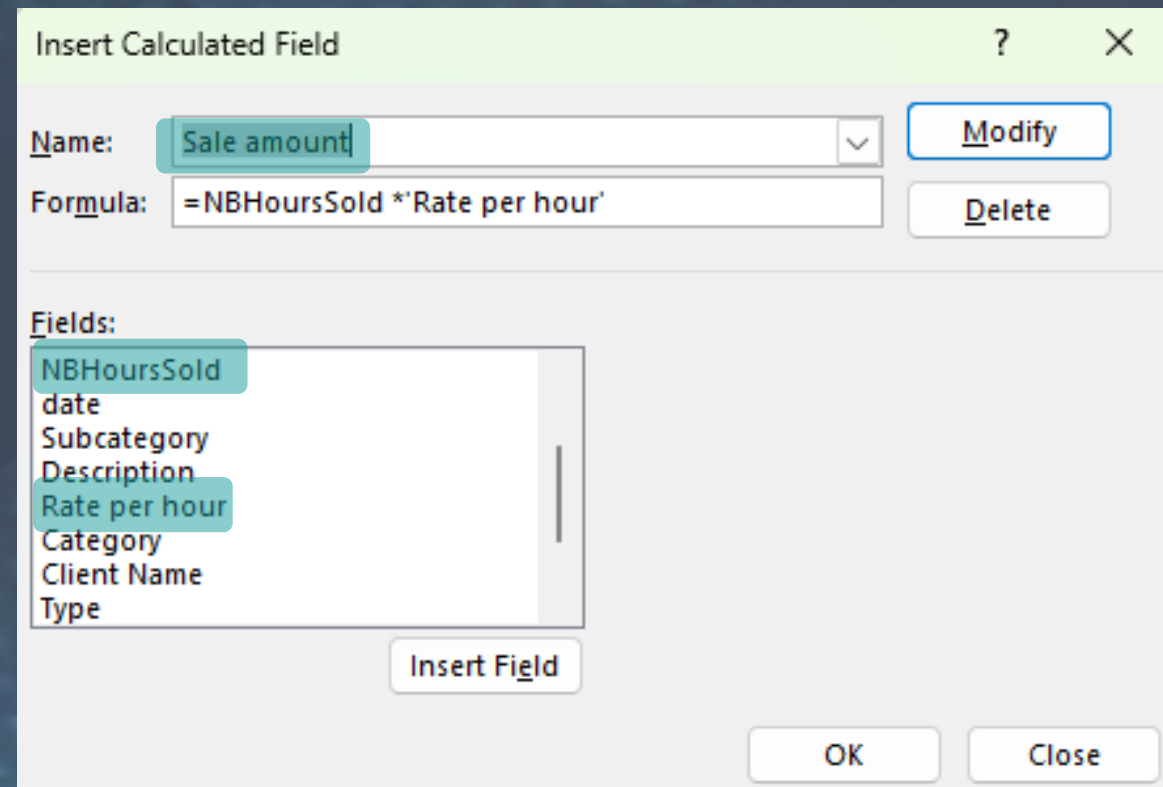
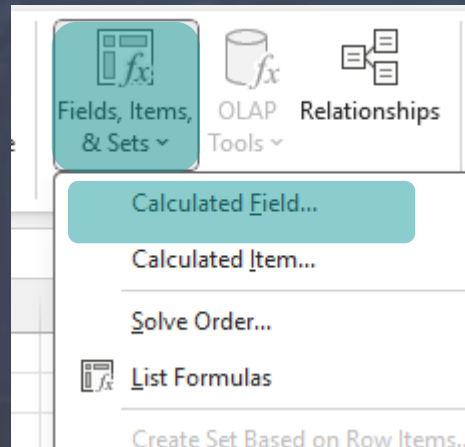
	A	B
1	Row Labels ▼	NB hours
2	BE	2417
3	BL	438
4	BS	323
5	FR	14761
6	GE	34159
7	JU	3024
8	LU	671
9	NE	8088
10	NW	461
11	SG	52
12	TI	3924
13	VD	30342
14	VS	10742
15	ZG	1355
16	ZH	764
17	Grand Total	111521

The screenshot shows the PivotTable Fields task pane. In the 'Rows' section, 'Canton' is selected. In the 'Values' section, 'NB hours' is selected. The task pane has a light gray background and a teal header bar.

Create a calculated field

GO to PivotTable Analyze > Fields, Items & Sets > Calculated Field

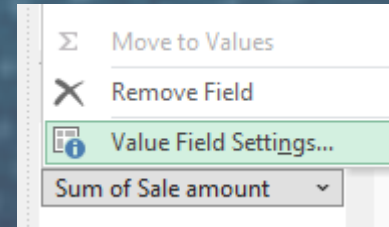
Type a name for this calculated field and create the formula by clicking the fields in the fields list separated by *.



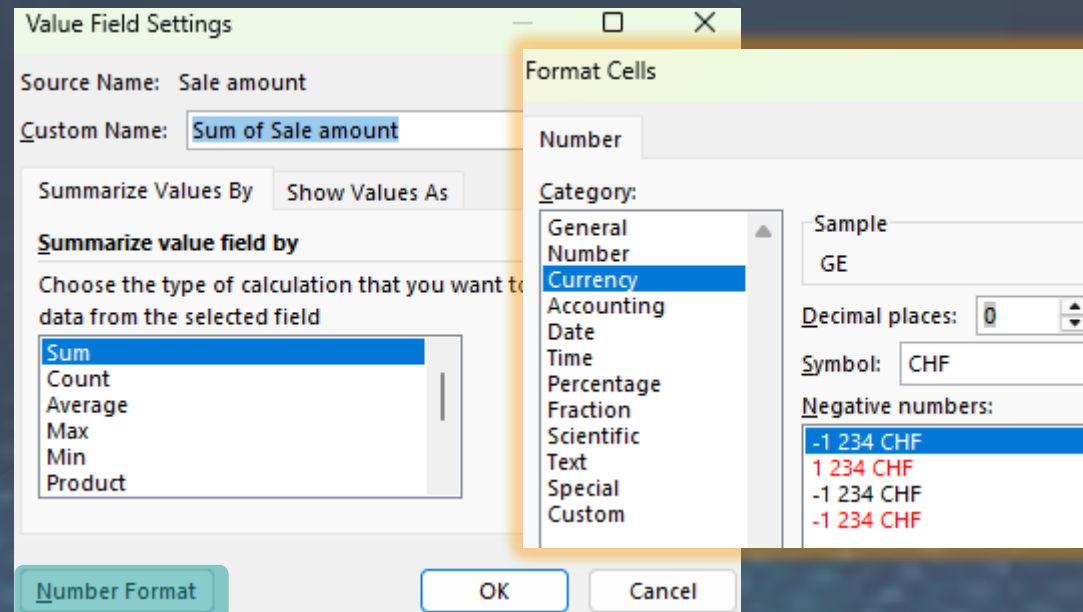
Create a calculated field

The Sale amount field is added to the pivot table.

Click on the drop down arrow on the right of the field in the Values window and choose 'Value Field Settings'.



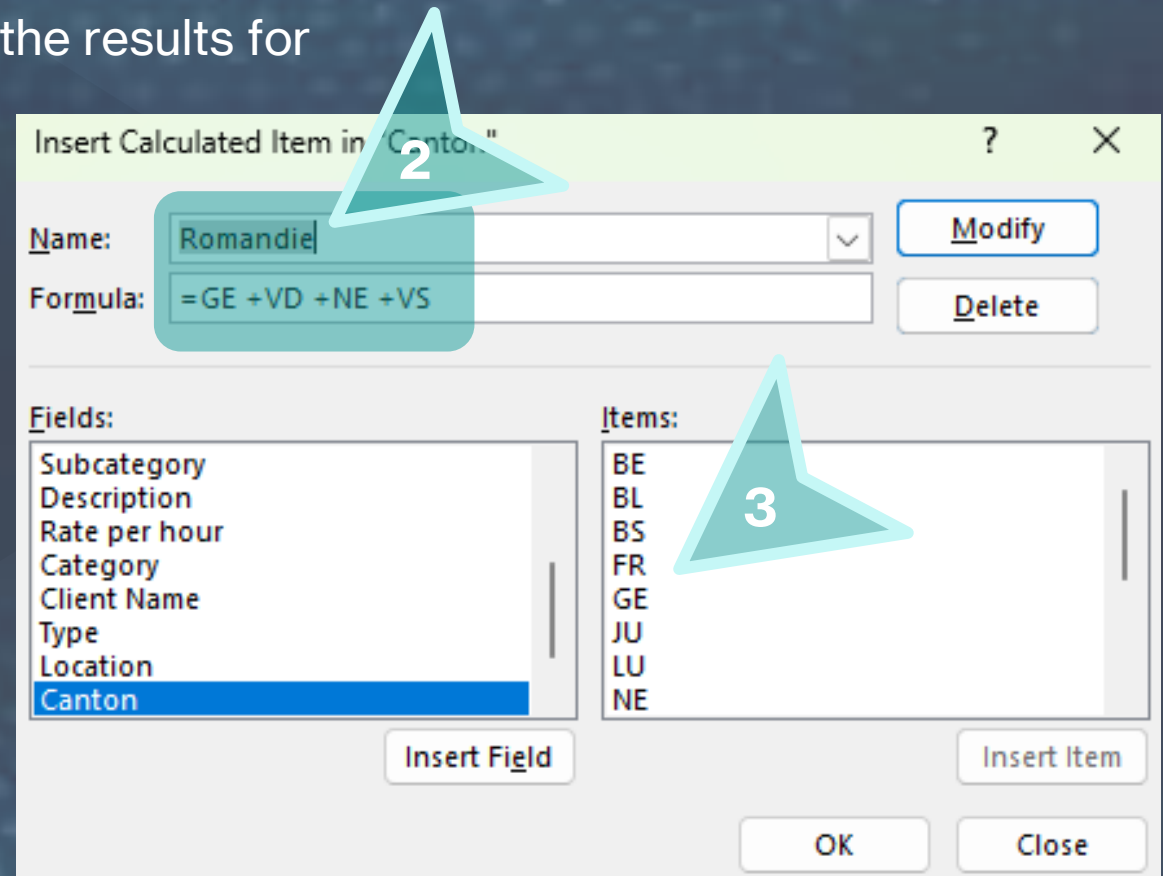
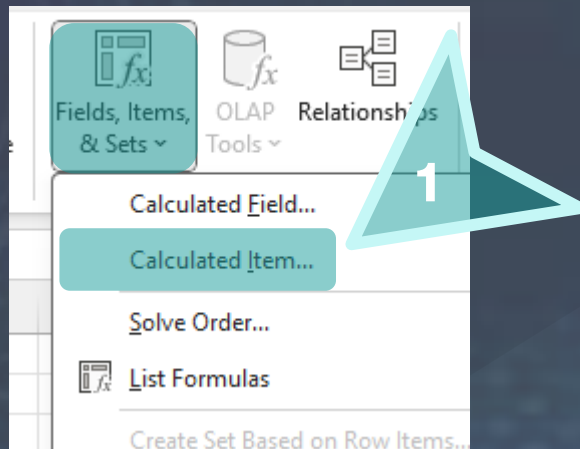
Select 'Number Format' in the popup window then format the field as currency with 0 decimal.



Create a calculated item

GO to PivotTable Analyze > Fields, Items & Sets > Calculated Item.

Type a name then create the formula that adds the results for several cantons.



Manual sort of the table

You can drag fields in the 1st column to reorder them within the pivot table

For example, drag 'Romandie' and 'VD' in the top 2 rows of the pivot table.

	A	B	C
1	Row Labels ▼	NB hours	Sum of Sale amount
2	Romandie	83331	11 564 676 180 CHF
3	VD	30342	1 586 583 180 CHF
4	BE	2417	9 160 430 CHF
5	BL	438	302 220 CHF
6	BS	323	177 650 CHF
7	FR	14761	350 868 970 CHF
8	GE	34159	1 943 305 510 CHF
9	JU	3024	12 216 960 CHF
10	LU	671	671 000 CHF
11	NE	8088	108 945 360 CHF
12	NW	461	322 700 CHF
13	SG	52	7 280 CHF
14	TI	3924	26 997 120 CHF
15	VS	10742	173 268 460 CHF
16	ZG	1355	2 533 850 CHF
17	ZH	764	1 230 040 CHF

Design of the pivot table

Copy the entire sheet.

In the rows window, replace the cantons by the Category and add the subcategory and the description to create an automatic hierarchy.

Use the Design tab of the pivot table to display / hide totals and subtotals. Add blank rows and choose the layout type.

Select your favorite color style.

Rename the pivot table and the sheet to Pvt_CatSubCatDesc. If not done yet, repeat for all current and future pivot tables.

	A	B	C
1	Row Labels	NB hours	Sum of Sale amount
2	Banking Services	5928	55 486 080 CHF
3	Business Banking	3592	20 689 920 CHF
4	Business Loans	2202	8 455 680 CHF
5	Merchant Services	1390	2 668 800 CHF
6			
7	Personal Banking	2336	8 409 600 CHF
8	Checking Accounts	1266	2 532 000 CHF
9	Savings Accounts	1070	1 712 000 CHF
10			
11	Credit & Debt	9732	126 516 000 CHF
12	Credit Counseling	4650	32 550 000 CHF
13	Credit Score Improvement	1088	1 523 200 CHF
14	Debt Settlement	3562	19 947 200 CHF
15			
16	Debt Management	5082	30 492 000 CHF
17	Credit Repair Services	2380	7 140 000 CHF
18	Debt Consolidation Plans	2702	8 106 000 CHF
19			
20	Estate Planning	25820	1 009 562 000 CHF
21	Inheritance Planning	11634	193 124 400 CHF
22	Beneficiary Designation	5538	38 766 000 CHF
23	Estate Tax Planning	6096	58 521 600 CHF
24			
25	Wills & Trusts	14186	319 185 000 CHF

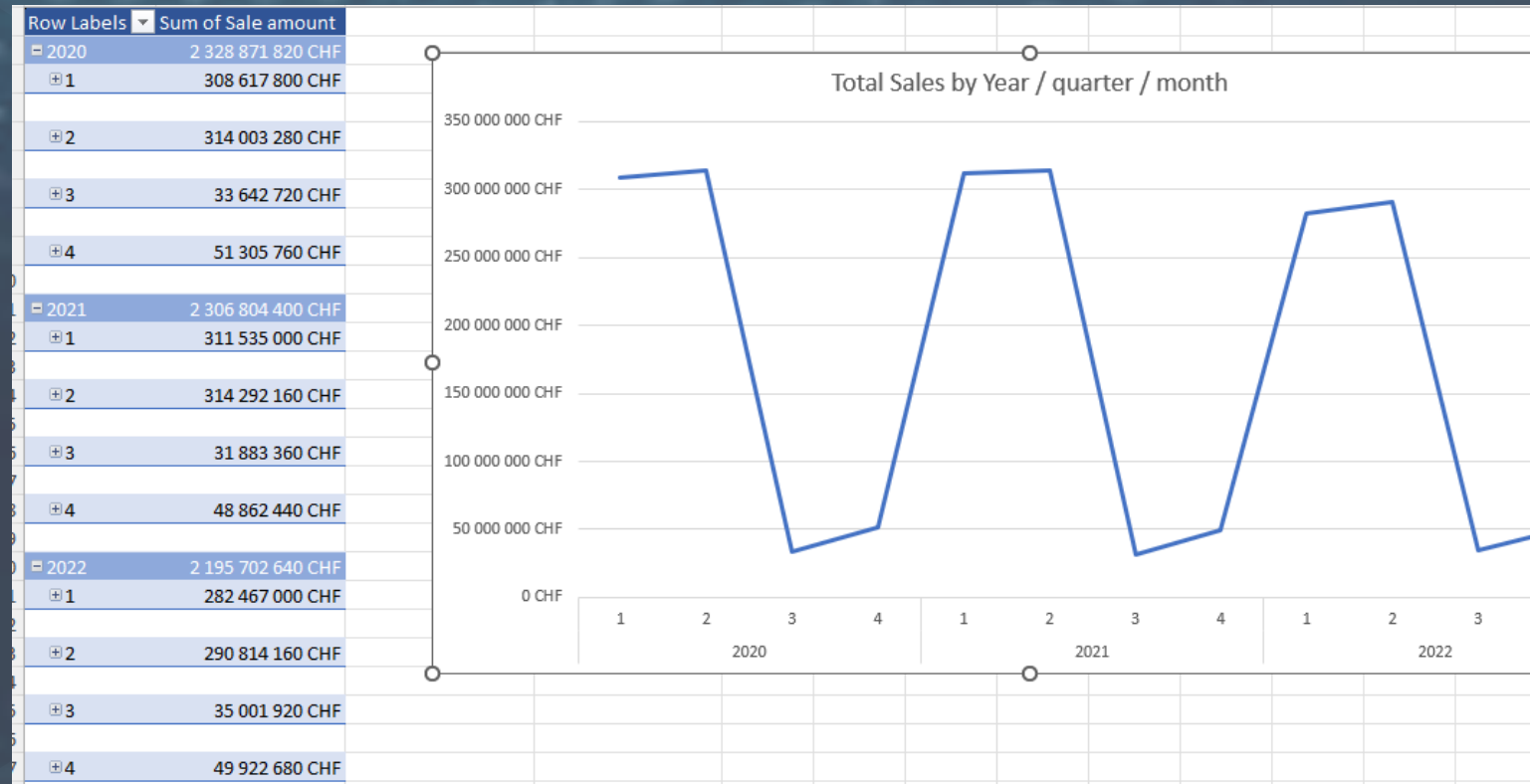
Create a date hierarchy

Copy the entire sheet.

Go to Power Query and in the factInvoices query, create the Year, quarter, Month and Month Name columns by adding date specific columns.

In the copied sheet, drag the Year, quarter and Month Name in the Rows. Keep only Sale amount in the values.

Insert a pivot chart (line chart) that displays to sales with the granularity given by the level in the date hierarchy.

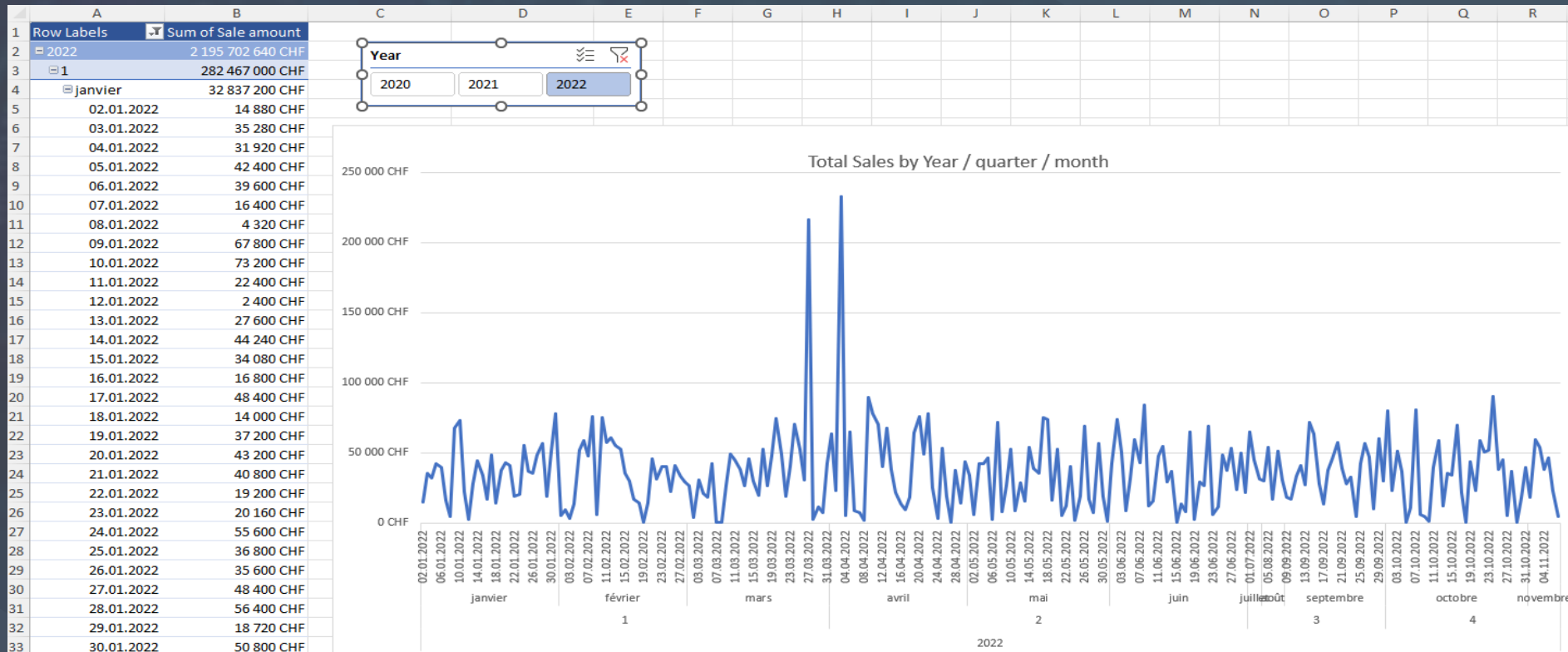


Analyze daily sales

Add the date field to the rows

Expand the date hierarchy to the date level.

Insert a slicer on Year and click on the desired year to view the data only for that specific year.



Show values as % and add cond. formatting

Copy the last sheet and drag the Category name in the rows and the Total Sales in the values.

Create the Total Sales field in Power Query by adding a custom column with the multiplication of the NBHoursSold by The Rate per Hour.

Custom Column

Add a column that is computed from the other columns.

New column name

Total sale

Custom column formula ⓘ

= [NBHoursSold] * [Rate per hour]

Available columns

- lineNB
- invoiceID
- clientID
- serviceID
- NBHoursSold
- date
- Description

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK Cancel

Show values as % and add cond. formatting

Insert the Total Sales field 5 times in the values window.

For the second, go to Field Value Settings, change the name and Choose 'Show Values As' and select '% of Column Total'. Format to percentage.

Value Field Settings

Source Name: Total sale

Custom Name: Sales %

Summarize Values By: Show Values As

Show values as

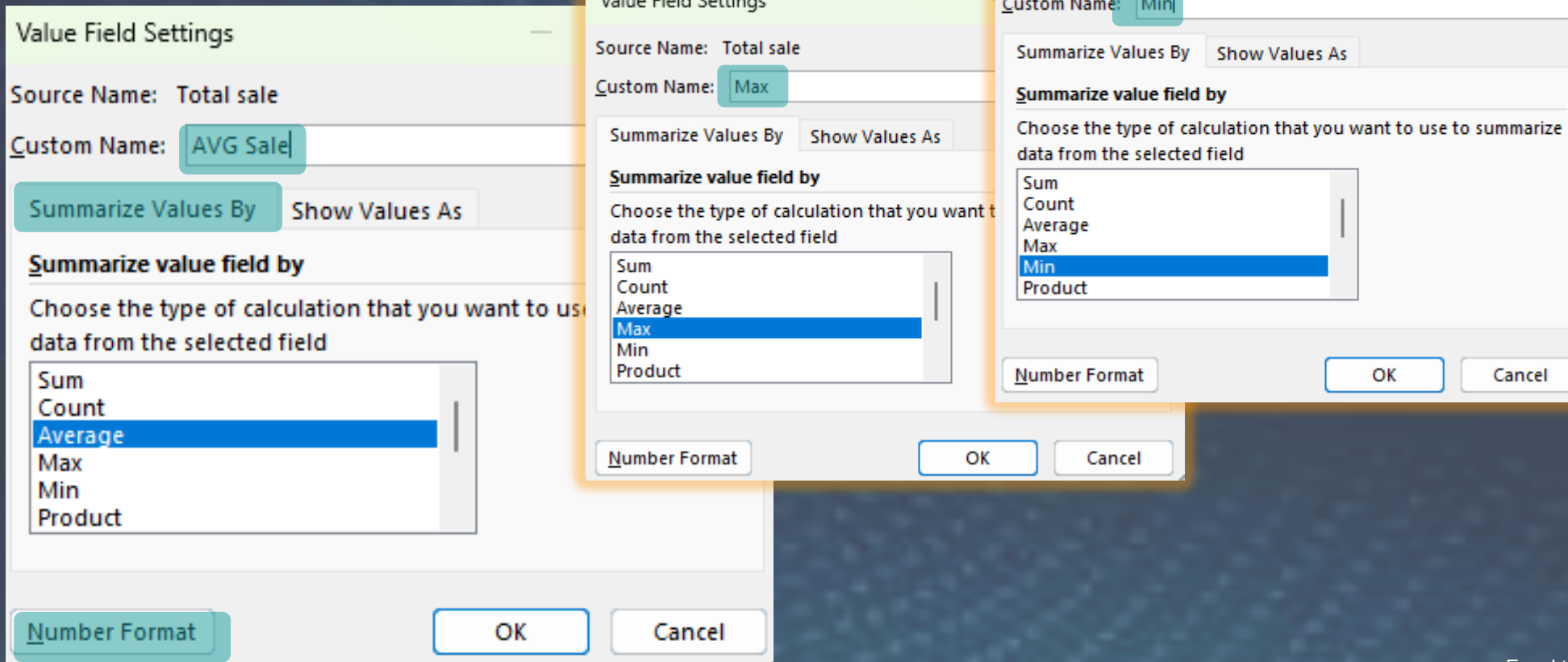
- % of Column Total
- No Calculation
- % of Grand Total
- % of Column Total
- % of Row Total
- % Of
- % of Parent Row Total

Type: Location

Number Format OK Cancel

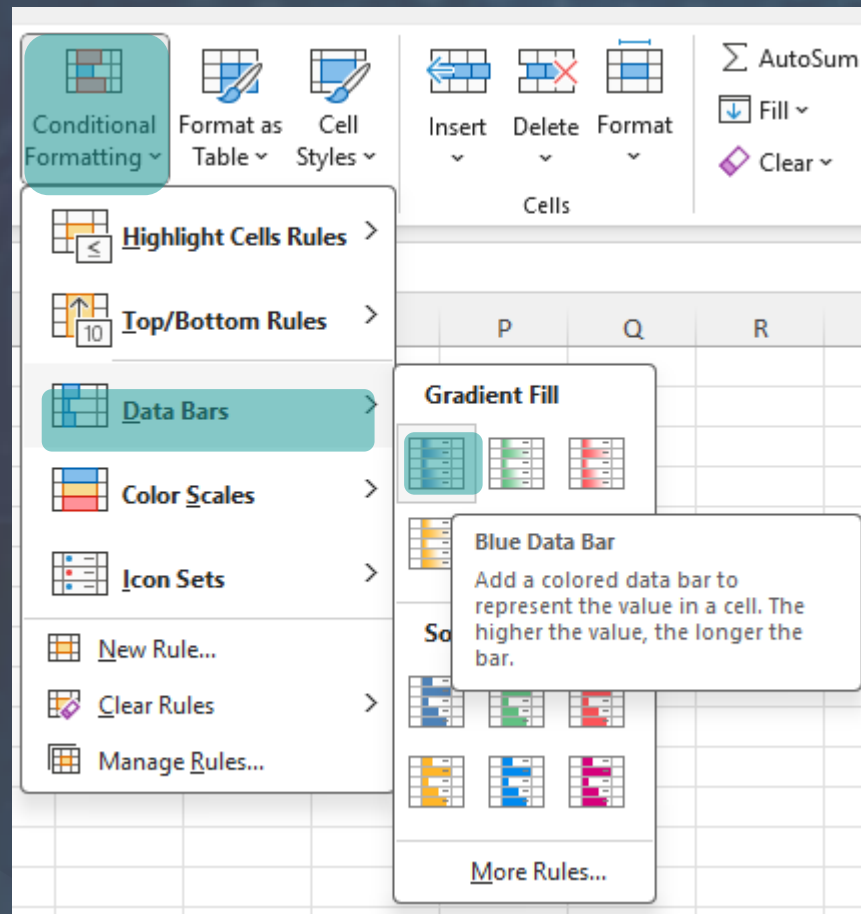
Show values as % and add cond. formatting

For the last 3 values of Total, change the summarization to average, maximum and minimum respectively.

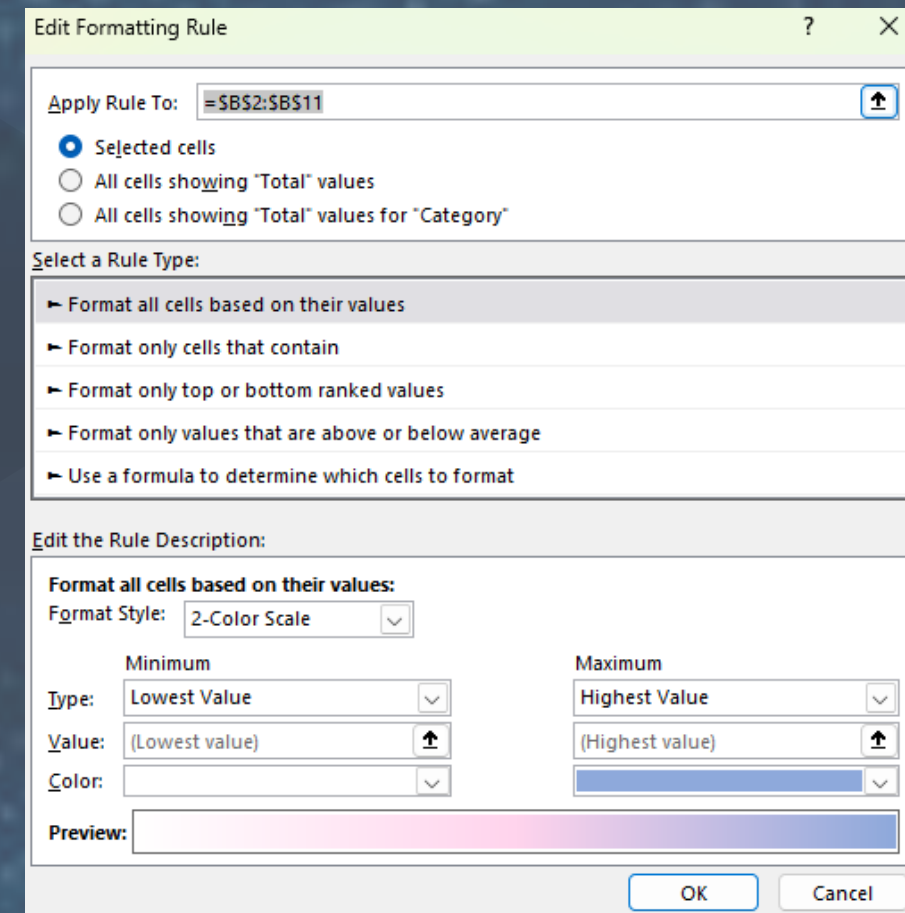


Show values as % and add cond. formatting

Apply conditional formatting to the Sales % column with a data bar.



Apply conditional formatting to the Total Sales column with a color scale and personalize the colors.



Final pivot table

The final Pivot Table should look like this. Insert a slicer on Year, Client type and Cantons to filter the results as you wish.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Row Labels	Total	Sales %	AVG sale	Max	Min		Canton					
2	Banking Services	664 640 CHF	4.82%	7 912 CHF	17760	360		BE	BL	BS			
3	Credit & Debt	973 200 CHF	7.06%	7 486 CHF	15000	400		FR	GE	JU			
4	Estate Planning	2 973 220 CHF	21.58%	8 797 CHF	21000	150		LU	NE	NW			
5	Financial Advisory	2 085 540 CHF	15.14%	12 414 CHF	27000	360		SG	TI	VD			
6	Financial Technology	1 321 360 CHF	9.59%	13 764 CHF	29200	420		VS	ZG	ZH			
7	Insurance Services	1 346 000 CHF	9.77%	8 108 CHF	17520	200							
8	Investment Services	775 080 CHF	5.63%	8 074 CHF	19600	100							
9	Mortgage Services	1 696 580 CHF	12.32%	7 747 CHF	17280	120							
10	Retirement Planning	640 800 CHF	4.65%	9 154 CHF	17520	120							
11	Tax Services	1 298 940 CHF	9.43%	10 647 CHF	21600	560							